

Geometry
Unit 9
Lesson 7 Practice

Name_____Answer Key_____
Date_____
Period_____

1. Austin lives by train tracks. He noticed a train goes by every afternoon. Most of the cars on the train are boxcars. A boxcar can be $60\frac{3}{4}$ feet long, $10\frac{2}{3}$ feet wide, and $11\frac{2}{3}$ feet tall. What is the maximum volume of a boxcar in the shape of a rectangular prism?

$$V = l \cdot w \cdot h = \left(60\frac{3}{4}\right)\left(10\frac{2}{3}\right)\left(11\frac{2}{3}\right)$$

$$V = 7560 \text{ ft}^3$$

2. Austin counted 21 boxcars on one train Saturday. If every boxcar was full, what was the total volume of cargo on the train?

$$21(7560) = 158,760 \text{ ft}^3$$

3. A ping pong ball has a diameter of 40 millimeters. What is the volume of a ping pong ball? Round to the nearest hundredth.

$$r = 20 \text{ mm}$$

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(20)^3$$

$$V \approx 33,510.32 \text{ mm}^3 \text{ (when using the pi button)}$$

4. The minimum diameter of a golf ball is 42.67 millimeters. What is the minimum volume of a golf ball? Round to the nearest hundredth.

$$r = 21.335 \text{ mm}$$

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(21.335)^3$$

$$V \approx 40,678.65 \text{ mm}^3 \text{ (when using the pi button)}$$

5. What is the difference in volume between a golf ball and a ping pong ball?

$$40678.65 - 33510.32 = 7168.33 \text{ mm}^3$$

6. Albert went on a road trip. During the drive, he noticed that some water towers were spherical, and some were cylindrical. He learned that the first water tower he saw was spherical with a 20 foot diameter. What was the volume of the first water tower? Round to the nearest hundredth.

$$r = 10$$

$$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(10)^3$$

$$V \approx 4188.79 \text{ ft}^3 \text{ (when using the pi button)}$$

7. The second water tower was cylindrical with a 19 foot diameter and a 21 foot height. What was the volume of the second water tower? Round to the nearest hundredth.

$$r = 9.5$$

$$V = \pi r^2 h = \pi(9.5)^2(21)$$

$$V \approx 5954.10 \text{ ft}^3 \text{ (when using the pi button)}$$

8. Which water tower can contain more water?

The cylindrical water tower can contain more water.

9. An Olympic swimming pool is 50 meters long, 25 meters wide, and 3 meters deep. What is the volume of water in a full pool?

$$V = lwh = (50)(25)(3)$$

$$V = 3750 \text{ m}^3$$

10. If the pool loses $\frac{1}{8}$ of its water depth during a free swim event, what is the new volume of water?

$$\text{Step 1) } 3750 \left(\frac{1}{8}\right) = 468.75$$

$$\text{Step 2) } 3750 - 468.75 = 3281.25 \text{ m}^3 \text{ left}$$